Mid Term Project Report

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Lesson Name: Internet and Web Programming

Lesson Number: SEN204

INTRODUCTION

Project Title:

Interactive Grocery Store Web Application

Research Problem / Problem Statement:

Managing grocery shopping efficiently can be challenging due to time constraints and lack of organization. Users often find it difficult to keep track of selected items and manage their purchases effectively while shopping.

Traditional methods do not provide a structured way to browse products or maintain a record of selected items. This project addresses the problem by providing a web-based solution that allows users to explore products and manage their shopping through a dynamic cart system, improving convenience and organization.

Objective of the Project:

The objective of this project is to develop an interactive and responsive grocery store web application using HTML, CSS, and JavaScript. The system allows users to browse products, interact with different sections, and manage their shopping efficiently.

It also demonstrates dynamic functionality using JavaScript, where user actions trigger real-time updates, ensuring a smooth and user-friendly experience across different devices.

Expected Outcome:

The expected outcome is a functional web application that enables users to browse products, add items to a cart, and view updates dynamically. The project showcases the use of JavaScript for interactivity and CSS for responsive design.

Overall, it demonstrates how web technologies can be used to create an effective and practical solution for grocery management.

METHODOLOGY

Approach to Problem Solving:

1. **Control Flow of the System:**

The system follows an event-driven approach where user actions trigger specific functions. When the user interacts with buttons such as search, cart, or add to cart, the system responds dynamically by updating the interface and data.

2. **Input, Processing, Output:**

Input:

- User clicks buttons (Add to Cart, Search, Menu, Login)
- User interacts with products and navigation links

Processing:

- JavaScript functions handle events
- Selected products are stored in an array data structure
- UI components are controlled using class toggling
- DOM manipulation updates the cart display dynamically

Output:

- Updated cart items displayed on screen
- UI panels (search, cart, login) open and close dynamically
- Smooth and responsive user experience

3. **System Control Flow Steps:**

1. User opens the website
2. Homepage is displayed
3. User navigates to products section
4. User clicks “Add to Cart”
5. Product is stored in an array
6. Cart display updates dynamically
7. User can view cart items
8. System continues until user stops interaction

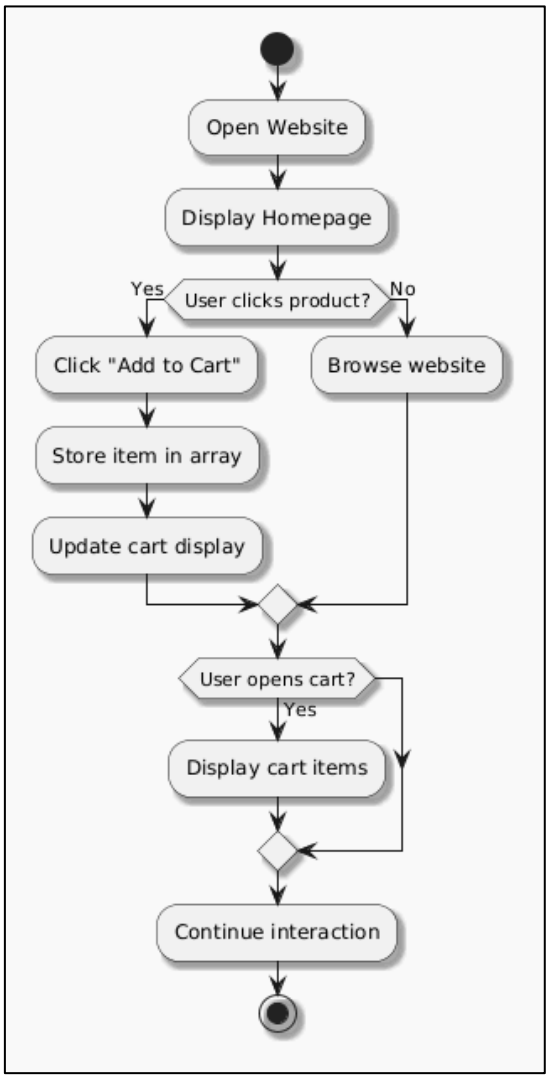

Figure 1: System Flowchart of Grocery Shopping Process

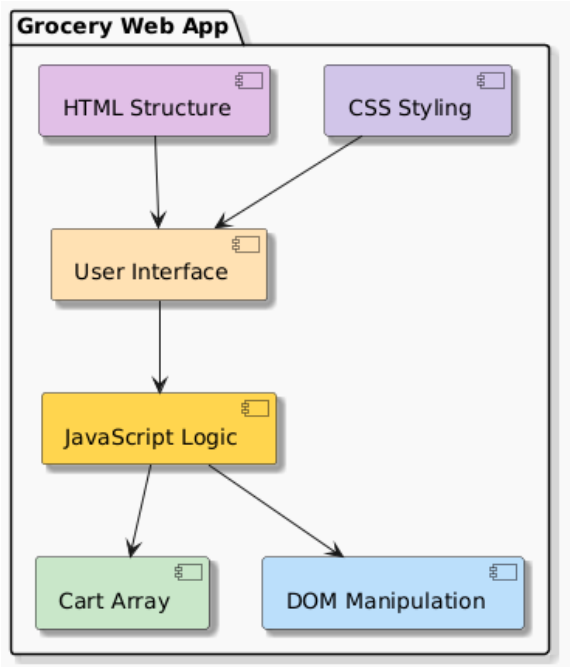

Figure 2: System Architecture and Component Interaction

TECHNICAL IMPLEMENTATION

Programming Languages:

HTML, CSS, JavaScript

Code Example:

```
  function message(){
    let msg = alert("Item Added to cart");
  }

  let cartItems = [];

function addToCart(productName, price) {
  const item = { productName, price };
  cartItems.push(item);
  displayCart();
}

function displayCart() {
  const cartList = document.getElementById('cart-items');
  cartList.innerHTML = '';
  cartItems.forEach(item => {
    const listItem = document.createElement('li');
    listItem.textContent = `${item.productName} - $${item.price}`;
    cartList.appendChild(listItem);
  });
}
```

The above figure presents a portion of the JavaScript code used in the application, as including the complete code is not feasible due to space limitations. This snippet highlights the core functionality of the system, where the addToCart() function stores selected products in an array (cartItems) as objects, representing the cart data structure. The displayCart() function then dynamically updates the user interface by manipulating the DOM, creating and displaying list elements for each item in the cart. This demonstrates the use of event-driven programming, data handling through arrays, and real-time interface updates, which are key aspects of the system's functionality.

Explanation:

The project is implemented using HTML for structure, CSS for styling and responsiveness, and JavaScript for interactivity and logic. The system uses event-driven programming where user actions trigger specific functions. When a user clicks the “Add to Cart” button, the product is stored in an array, and the cart is updated dynamically using DOM manipulation. The interface also uses class-based toggling to control visibility of search, login, and cart sections. Additionally, responsive design techniques ensure the system works across different devices.

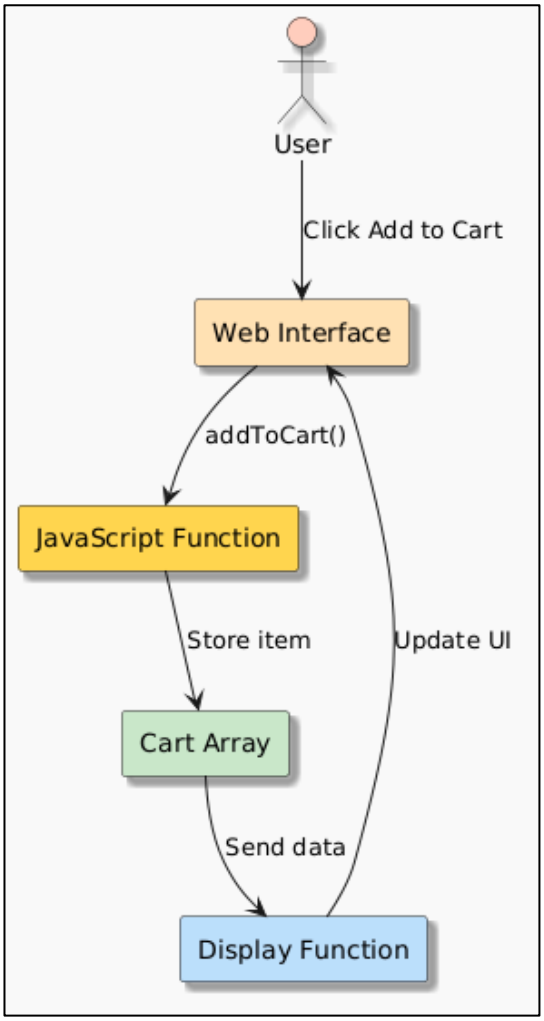

Figure 3: Data Flow in Grocery Cart System